Topological complexity two in the homogeneous lower model

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Abstract

In the lower multiparameter model of random simplicial complexes, a simplex is retained when all its nonempty subfaces pass independent Bernoulli trials. We prove that, when every trial has the same fixed probability in $(0,1)$, the resulting complex is path-connected and has reduced topological complexity two with probability tending to one along all ambient sample sizes. Thus three continuous local motion-planning rules are necessary and sufficient. This removes the density-one restriction in the result of Barmak and Farber. The new step is an anti-concentration estimate for the Euler characteristic: after the other variables are exposed, many remaining face trials independently add or remove a single maximal simplex.

1 Statement and topological reduction

Topological complexity measures the number of continuous local rules needed to plan paths between pairs of points. We use its reduced convention: for a path-connected space Y, $\mathrm{TC}(Y)$ is the least $k \geq 0$ such that $Y \times Y$ admits an open cover by $k+1$ sets, each with a continuous local section of

$$Y^{[0,1]} \longrightarrow Y \times Y, \qquad \gamma \longmapsto (\gamma(0), \gamma(1)).$$

Fix $p \in (0,1)$. For each nonempty $\sigma \subseteq [n] = \{1, \ldots, n\}$, let ξ_σ be an independent Bernoulli variable with success probability p, and define

$$X_n = \{\varnothing \neq \sigma \subseteq [n] : \xi_\tau = 1 \text{ for every } \varnothing \neq \tau \subseteq \sigma\}. \tag{1}$$

Vertices are sampled as well as higher-dimensional faces. This is the homogeneous medial regime of the lower model studied in [1]. We distinguish the ambient sample size n from the random number of retained vertices.

Farber proposed the value two for medial-regime random complexes in [2, Remark 8.2]. Barmak and Farber proved the upper bound $\mathrm{TC} \leq 2$ along all sample sizes and equality along a subset of integers of natural density one [1, Theorems 16 and 19]. Their lower bound uses a face number that exceeds its two neighbouring face numbers. Near transitions where adjacent counts are comparable, this argument does not supply the lower bound. They ask whether the sample-size restriction can be removed [1, §10, Question 2].

Theorem 1. *For every fixed $p \in (0,1)$, the complex (1) satisfies*

$$\mathbb{P}\{|X_n| \text{ is path-connected and } \mathrm{TC}(|X_n|) = 2\} \longrightarrow 1 \quad \text{as } n \longrightarrow \infty$$

through all positive integers.

The upper bound and connectivity are inherited from [1]. We obtain the lower bound by showing that the Euler characteristic has no point mass of asymptotically positive size. This avoids choosing a dimension in which homology must dominate. All limits below hold for fixed p; constants and eventual thresholds may depend on p.

For a finite simplicial complex K, write

$$\chi(K) = \sum_{\sigma \in K} (-1)^{|\sigma|-1}, \qquad b_j(K) = \dim_{\mathbb{Q}} H_j(K; \mathbb{Q}).$$

The next observation explains the role of the Euler characteristic.

Lemma 2. *If K has path-connected realization and $\chi(K) \notin \{0, 1\}$, then $\mathrm{TC}(|K|) \geq 2$.*

Proof. By [1, Lemma 17], each of the following implies $\mathrm{TC} \geq 2$: a nonzero Betti number in a positive even degree, an odd-degree Betti number at least two, or nonzero Betti numbers in two distinct positive odd degrees. If none occurs, $b_0 = 1$, every positive even Betti number vanishes, and the sum of the odd Betti numbers is at most one. The Euler–Poincare identity then gives $\chi(K) \in \{0, 1\}$. Taking the contrapositive proves the lemma. □

2 Anti-concentration of the Euler characteristic

Theorem 3. *For the complex (1) with fixed $p \in (0, 1)$,*

$$\sup_{z \in \mathbb{Z}} \mathbb{P}\{\chi(X_n) = z\} \longrightarrow 0 \quad \text{as } n \longrightarrow \infty \tag{2}$$

through all positive integers.

We select many faces whose raw variables will remain unexposed. They will have successful boundaries and blocked cofaces, so each remaining trial changes the Euler characteristic by exactly one, with a common sign. The separation of the selected faces ensures that the coface tests do not examine any of their raw variables.

2.1 A separated family of faces

Put $q = 1/p$ and, for sufficiently large n, choose

$$r = \lceil \log_2(2 \log_q n) \rceil. \tag{3}$$

Here r is the number of vertices of a selected simplex, not its dimension. This level makes cofaces rare while leaving many successful boundaries. Indeed,

$$2 \log_q n \leq 2^r < 4 \log_q n, \qquad n^{-4} < p^{2^r} \leq n^{-2}. \tag{4}$$

Also $r \to \infty$ and $r = O_p(\log \log n)$. We work beyond a threshold where $r \geq 6$ and $n \geq 4r$; only finitely many initial sample sizes are therefore omitted from the estimates.

Colour every r-element subset of $[n]$ by the sum of its vertices modulo n, and choose a largest colour class F. Resolve ties by the smallest residue. If $m = |F|$ and $N = \binom{n}{r}$, then $m \geq N/n$. Two distinct members of F cannot share $r - 1$ vertices: their sums would differ by two distinct labels, whose nonzero difference has absolute value less than n. Consequently every $(r + 1)$-element set contains at most one member of F.

For fixed $\sigma \in F$, let $M_i(\sigma)$ count the members $\tau \in F$ with $|\sigma \cap \tau| = i$. For $0 \leq i \leq r - 2$,

$$M_i(\sigma) \leq \binom{r}{i} \frac{\binom{n-r}{r-i-1}}{r-i} = \frac{\binom{r}{i}\binom{n-r}{r-i}}{n - 2r + i + 1}. \tag{5}$$

To prove this, choose the intersection and put $s = r - i$. For an $(s-1)$-subset outside σ, the required residue determines the missing vertex uniquely in $[n]$. Each admissible s-subset is counted s times by deleting one element; inadmissible completions only decrease the count. This argument applies also to composite n.

2.2 Successful boundaries

For $\sigma \in F$, let

$$A_\sigma = \prod_{\varnothing \neq \tau \subsetneq \sigma} \xi_\tau, \qquad a = p^{2^r - 2}, \qquad W = \sum_{\sigma \in F} A_\sigma, \qquad \mu = \mathbb{E}W = ma.$$

Thus A_σ records whether the entire proper boundary succeeds, including all singleton variables. None of the variables indexed by F has been used. Since $\binom{n}{r} \geq (n/r)^r$,

$$\mu \geq p^{-2} n^{r-5} / r^r \longrightarrow \infty. \tag{6}$$

For the limit, $\log r / \log n \to 0$ and $r \to \infty$, so $(r-5)\log n - r \log r \to \infty$.

Distinct $\sigma, \tau \in F$ with intersection size i share exactly $2^i - 1$ proper-boundary variables. Hence

$$\mathbb{E}(A_\sigma A_\tau) = a^2 q^{2^i - 1}.$$

Disjoint pairs have zero covariance, and pairs with $i = r - 1$ do not occur. Set

$$H_i = \frac{\binom{r}{i}\binom{n-r}{r-i}}{\binom{n}{r}}.$$

By (5), $M_i(\sigma)/m \leq 2H_i$, because $n/(n - 2r + i + 1) \leq 2$. Summing ordered pairs and the diagonal gives

$$v_n := \frac{\operatorname{Var} W}{\mu^2} \leq \frac{1}{\mu} + 2 \sum_{i=1}^{r-2} H_i (q^{2^i - 1} - 1). \tag{7}$$

We have

$$H_i \leq \left(\frac{r^2}{n - r + 1} \right)^i \leq (2r^2/n)^i. \tag{8}$$

For the first inequality, replace $\binom{n-r}{r-i}$ by $\binom{n-i}{r-i}$, whose ratio to $\binom{n}{r}$ is $(r)_i/(n)_i$, where $(u)_i = u(u-1)\cdots(u-i+1)$. Then use $\binom{r}{i} \leq r^i$ and $(r)_i/(n)_i \leq [r/(n-r+1)]^i$.

The sequence $2^i/i$ increases for $i \geq 2$. Thus, uniformly for $2 \leq i \leq r-2$,

$$4\, 2^{i-r} \leq \frac{i}{r-2} \leq \frac{i}{2}, \qquad q^{2^i} \leq n^{4\, 2^{i-r}} \leq n^{i/2},$$

where (4) supplies $q^{2^r} < n^4$. Put $y = 2r^2/\sqrt{n} \to 0$. For sufficiently large n, $y < 1$, and (7)–(8) yield

$$v_n \leq \frac{1}{\mu} + \frac{4(q-1)r^2}{n} + \frac{2y^2}{1-y} \longrightarrow 0. \tag{9}$$

This controls the whole changing overlap range, not only each fixed i. Chebyshev's inequality now gives $\mathbb{P}\{W < \mu/2\} \leq 4v_n$.

2.3 Cofaces and conditioning

Let $\mathcal{G}$ be the sigma-field generated by all raw variables whose indices are outside the deterministic family F. The successful-boundary indicators are $\mathcal{G}$-measurable. For $\sigma \in F$ and $v \notin \sigma$, set $\rho = \sigma \cup \{v\}$ and

$$U_{\sigma,v} = \prod_{\substack{\varnothing \neq \tau \subseteq \rho \\ \tau \neq \sigma}} \xi_\tau.$$

This product examines every variable needed to retain ρ except ξ_σ. It is $\mathcal{G}$-measurable: ρ contains at most one member of F, and that member is σ.

Let T count the members of F having successful boundaries and at least one v for which $U_{\sigma,v} = 1$. Given $A_\sigma = 1$, a fixed coface requires exactly 2^r new independent variables, namely the subsets containing v, including ρ itself. Therefore a union bound gives

$$\mathbb{E}T \leq ma(n-r)p^{2^r} \leq \mu/n. \tag{10}$$

Define

$$S = \{\sigma \in F : A_\sigma = 1,\ U_{\sigma,v} = 0 \text{ for every } v \notin \sigma\}, \qquad K = |S| = W - T.$$

The set S is determined by $\mathcal{G}$. By Markov's and Chebyshev's inequalities,

$$\mathbb{P}\{K < \mu/4\} \leq 4v_n + 4/n \longrightarrow 0. \tag{11}$$

Indeed $W \geq \mu/2$ and $T \leq \mu/4$ imply $K \geq \mu/4$.

For $\sigma \in S$, its proper subfaces are present and all immediate cofaces are blocked by failed variables outside F. They remain absent for every completion of the unexposed variables. A larger coface would contain an immediate coface, so is absent as well. Changing ξ_σ therefore changes membership of σ alone, and changes the Euler characteristic by $(-1)^{r-1}$.

Condition on $\mathcal{G}$, then expose the variables indexed by $F \setminus S$. For each positive-probability $\mathcal{G}$-atom the set S is fixed. The original product law still factors on the unexposed coordinates, so these K variables are independent Bernoulli(p) trials. Thus, conditionally,

$$\chi(X_n) = C + (-1)^{r-1}B_K, \qquad B_K \sim \mathrm{Bin}(K,p), \tag{12}$$

where C is fixed by the exposed data. The choice of S has not used any coordinate that remains unexposed.

2.4 Point masses

For $B_k \sim \mathrm{Bin}(k,p)$ with $k \geq 1$ and any integer j, expanding the finite polynomial $(1-p+pe^{it})^k$ and integrating its integer frequencies gives

$$\mathbb{P}\{B_k = j\} = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-ijt}(1-p+pe^{it})^k\,dt.$$

For $|t| \leq \pi$,

$$|1-p+pe^{it}|^2 = 1 - 4p(1-p)\sin^2(t/2), \qquad \sin(|t|/2) \geq |t|/\pi.$$

Using $1-u \leq e^{-u}$ and extending a positive Gaussian integral, whose value follows from [3, §7.2(i)], we obtain

$$\begin{aligned}\sup_j \mathbb{P}\{B_k = j\} &\leq \frac{1}{2\pi}\int_{-\pi}^{\pi} \exp[-2kp(1-p)t^2/\pi^2]\,dt \\ &\leq \frac{\sqrt{\pi}}{2\sqrt{2kp(1-p)}} = \frac{C_p}{\sqrt{k}}, \qquad C_p = \frac{\sqrt{\pi}}{2\sqrt{2p(1-p)}}.\end{aligned} \tag{13}$$

Since $\mu \to \infty$, the event $K \geq \mu/4$ eventually ensures $K \geq 1$. On that event, (12)–(13) bound every conditional point mass of $\chi(X_n)$ by $2C_p/\sqrt{\mu}$. Use the bound one on its complement and (11) to conclude

$$\sup_{z\in\mathbb{Z}} \mathbb{P}\{\chi(X_n) = z\} \leq 4v_n + 4/n + 2C_p/\sqrt{\mu} \longrightarrow 0.$$

This proves Theorem 3.

3 Proof of the topological complexity statement

Proof of Theorem 1. For every simplicial subcomplex L on $[n]$, (1) retains all its simplices exactly when their raw Bernoulli(p) trials succeed, because L contains every required nonempty subface. Independence gives $\mathbb{P}\{L \subseteq X_n\} = p^{\#L}$, where $\#L$ is the number of nonempty faces of L. Thus X_n has precisely the homogeneous lower-model law of [1, §3.1 and §6.1], including random vertices. Its parameters $p_\sigma = p$ satisfy the fixed bounds

$$0 < p_- = p/2 \leq p_\sigma \leq p_+ = (1+p)/2 < 1,$$

which are the medial-regime hypotheses. The uniform positive lower bound also satisfies the hypothesis of Corollary 10 of [1], which gives path connectivity with probability tending to one along all n, since its connectivity level tends to infinity. Theorem 16 of the same source gives the reduced upper bound $\mathrm{TC} \leq 2$ with probability tending to one along all n.

By Theorem 3, the probability of $\chi(X_n) \in \{0, 1\}$ tends to zero. On the intersection of the connectivity event, the upper-bound event, and $\chi(X_n) \notin \{0, 1\}$, Lemma 2 gives $\mathrm{TC} \geq 2$. This intersection has probability tending to one, proving the theorem. □

References

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[3] NIST Digital Library of Mathematical Functions, §7.2(i), equations 7.2.1 and 7.2.2. https://dlmf.nist.gov/7.2. Accessed 14 October 2026.

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